

Technical Proposal: Modules 7 & 8 Revision

M7: Forum & Discussion (Revised)

A community feature allowing users to discuss their job application process, share company interview insights, and seek peer advice. Students can create a post and assign it categorization tags/labels directly within the creation interface. Students can like other users' posts, add comments, or delete their own contributions.

The forum page is split into two sections: a primary feed sorted by engagement (likes/recency) featuring an integrated text search bar, and a sidebar displaying active taxonomy tags to filter posts.

Database Schema

posts

Column	Type	Constraints
post_id	INT	PK, AUTO_INCREMENT
user_id	INT	FK users(user_id)
title	VARCHAR(150)	NOT NULL
content	TEXT	NOT NULL
tag	VARCHAR(60)	DEFAULT 'General'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

comments

Column	Type	Constraints
comment_id	INT	PK, AUTO_INCREMENT
post_id	INT	FK posts(post_id) ON DELETE CASCADE
user_id	INT	FK users(user_id)

Column	Type	Constraints
content	TEXT	NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

post_likes

Column	Type	Constraints
like_id	INT	PK, AUTO_INCREMENT
post_id	INT	FK posts(post_id) ON DELETE CASCADE
user_id	INT	FK users(user_id)
UNIQUE(post_id, user_id)		

API Endpoints (M7)

Method	Path	Auth	Action
GET	/api/forums	Any logged-in	List all posts (title, tag, like/comment count, author)
GET	/api/forums/{id}	Any logged-in	View a single post and its structural comments
POST	/api/forums	Student	Create a forum post (includes an optional tag string)
PUT	/api/forums/{id}	Student (Owner)	Edit own post text or tag content

Method	Path	Auth	Action
DELETE	/api/forums/{id}	Student / Admin	Delete forum post (cascades linked comments/likes)
POST	/api/forums/{id}/comments	Student	Add a comment thread entry
DELETE	/api/forums/comments/{id}	Student / Admin	Delete a specific comment entry
POST	/api/forums/{id}/like	Student	Toggle like status on an entry

M8: Mock Interview & Technical Prep Scheduler

A robust system designed to help students prepare for live technical interviews with dedicated career administrators. This module manages scheduling allocations and tracks formal mock evaluation history using full CRUD functionality across relational dependencies.

Admin Operations:

- Create open calendar availability slots specifying a target datetime block.
- Read all global schedules, booked sessions, and track historical results.
- Update pending scheduled slots to append test scores and structured text feedback after an interview concludes.
- Delete empty, unbooked calendar slots if availability windows shift.

User (Student) Operations:

- Read all available unbooked administrative slots filtered by timing.
- Create a reservation booking mapped to an open slot, identifying their target technical domain (e.g., Software Engineering, Data Science, UI/UX).
- Update or reschedule their chosen domain or cancel a reservation before it takes place.
- Read their personalized historical dashboard containing past interview scores and written performance feedback.

Database Schema

interview_slots

Column	Type	Constraints
slot_id	INT	PK, AUTO_INCREMENT
interviewer_id	INT	FK users(user_id)
scheduled_at	DATETIME	NOT NULL
is_booked	BOOLEAN	DEFAULT FALSE

mock_interviews

Column	Type	Constraints
interview_id	INT	PK, AUTO_INCREMENT
slot_id	INT	FK interview_slots(slot_id) ON DELETE CASCADE
student_id	INT	FK users(user_id)
job_category	VARCHAR(100)	NOT NULL (e.g., Software Engineering, Design)
status	ENUM	'pending', 'completed', 'cancelled' DEFAULT 'pending'
feedback_text	TEXT	NULLABLE
score	INT	NULLABLE

API Endpoints (M8)

Method	Path	Auth	Action
GET	/api/interviews/slots	Any	List all available open time slots

Method	Path	Auth	Action
		logged-in	
POST	/api/interviews/slots	Admin	Create a new open availability time slot
DELETE	/api/interviews/slots/{id}	Admin	Delete an unbooked time slot
GET	/api/interviews/my-sessions	Student	View a student's personal bookings and scores
POST	/api/interviews/bookings	Student	Book an available slot (creates interview tracking entry)
PUT	/api/interviews/bookings/{id}	Student	Update booking preferences or set status to cancelled
GET	/api/interviews/admin/manage	Admin	Read all global student bookings and interview statuses
PUT	/api/interviews/admin/evaluate/{id}	Admin	Complete interview; submit formal score and feedback text

Technical Proposal: Modules 7 & 8 Revision

M7 — Forum & Discussion

A new feature allowing users to discuss their job applications process, the companies available, etc.

Students can create post with a specific label, which they either choose from existing labels or create a new one. Students can like the posts of other users or add a comment for them. They can delete their previous comments or posts.

The page will be divided into 2 sections. Posts will be displayed on one side and sorted based on how many likes they have. The other side will be reserved for labels that have posts under them.

Users can use a search bar at the top to search between posts or click on the labels to view only the posts under this label.

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Column	Type	Constraints
		CASCADE
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post_likes

Column	Type	Constraints
like_id	INT	PK, AUTO_INCREMENT
post_id	INT	FK posts(post_id) ON DELETE CASCADE
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